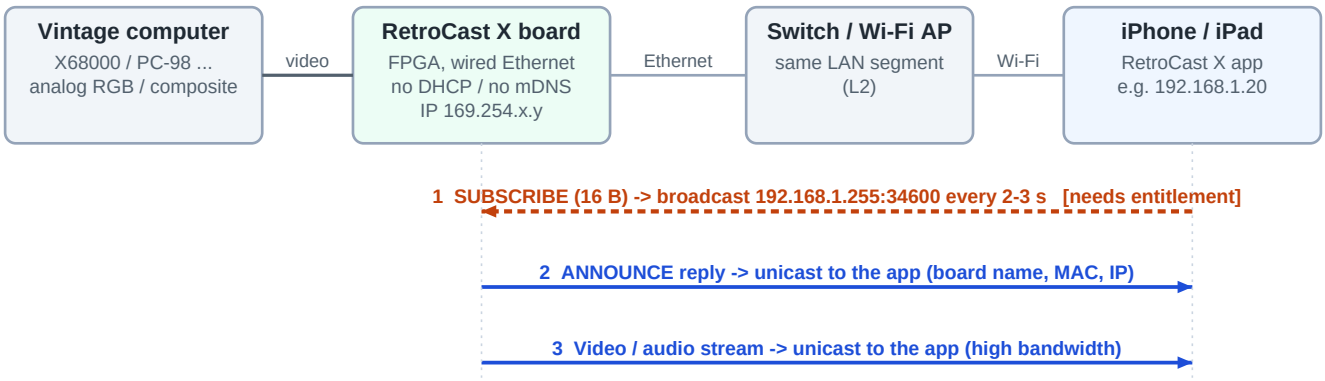
RetroCast X - Request for the Multicast Networking entitlement

App: RetroCast X | Bundle IDs: jp.ohnaka.RetroCastX, jp.ohnaka.RetroCastX.dev | Open source: github.com/kunichiko/RetroCastX

What the app does

RetroCast X is a viewer for a hobbyist hardware device of the same name: an FPGA board that captures the analog video of vintage computers (Sharp X68000, NEC PC-98 and others) and streams it uncompressed over wired Ethernet to a viewer app on the same LAN. The macOS and Windows viewers are already in use; the iOS / iPadOS version is built from the same code.

Network flow



Only step 1 is a broadcast. No multicast group is joined, and the app never needs to receive a broadcast: every packet from the board, including the video stream, is unicast to the app.

Why broadcast is required

- The board implements a minimal UDP/IP stack in FPGA logic. It has **no DHCP client and no mDNS / Bonjour**.
- By default it uses a **link-local address derived from its MAC** (169.254.x.y), so it is normally on a **different subnet** from the iPhone/iPad. Unicast discovery is not possible without asking the user to type an IP address.
- The board accepts broadcasts regardless of subnet and replies by unicast to the sender's MAC/IP. This works today on macOS, Windows and Android.

Packets the app sends by broadcast

Message	Size	When	Content
SUBSCRIBE (discovery)	16 bytes	every 3 s	8-byte header (magic 0x52 'R', version 0, type 4, flags bit0 = announce-only, 16-bit frame / sequence) + target MAC (ff:ff:ff:ff:ff:ff = any board) + 2 reserved bytes. Each board replies with a unicast ANNOUNCE.
SUBSCRIBE (keep-alive)	16 bytes	every 2 s while viewing	Same layout, flags bit0 = 0, target MAC = the selected board. Only that board reacts and keeps streaming to the app; the subscription expires after 10 s without it.
CONFIG	24 bytes	only when the user changes a setting	Header + target MAC + register key (u16) + value (u32). The board replies by unicast.

Transport: UDP over IPv4, port 34600 (source and destination). Destination is the subnet-directed broadcast address of each active interface (e.g. 192.168.1.255); 255.255.255.255 is not used. **Traffic:** one 16-byte packet every 2-3 seconds, plus rare 24-byte CONFIG packets.

Protocol specification: https://github.com/kunichiko/RetroCastX/blob/main/docs/protocol-v0.md